

1. Reverse a String without slicing:

```
str = input("Enter a string: ")
rev = ""
for i in range (len(str)-1,-1,-1):
    rev += str[i]
print(rev)
```

1. Reverse a String with slicing:

```
str = input("Enter a string: ")
revStr = str[::-1]
print(revStr)
```

2. Count words in a sentence:

```
str = input("Enter a string: ")
count = 1
for i in str:
    if i==" ":
        count+=1
print("No of words is ",count)
```

3. Remove duplicate characters:

```
str = input("Enter a string: ")
newStr=""
for i in range (0,len(str)):
    if str[i]==str[i-1]:
        newStr=str[:i]+str[i+1:]
print(newStr)
```

4. Difference between lists and sets:

Lists:

1. List have ordered elements.
2. List can have duplicate elements.
3. List can contain multiple data types including list.
4. List is indexed.

Sets:

1. Set have unordered elements.
2. Set cannot have duplicate elements.
3. Set can only contain integers, strings and tuples.
4. Set is unindexed.

5. Can sets contain lists:

No sets cannot contain lists as sets use a hash table to store values. Since values in hash table cannot be changed after they are stored and lists are mutable, lists cannot be stored in sets.

6. Why sets are faster:

Sets are faster because they use a hash table to store and read values. Since the hash table only contains unique values, sets are faster compared to other data types.

7. Difference between `remove()` and `discard()`:

`remove()`:

1. Gives an error if the element to be removed is missing.

`discard()`:

1. Does not give an error if the element to be removed is missing.

8. How many methods to insert element in a list. Total methods:

There are 3 methods to insert element in a list. They are:

1. `append()`: Adds a single element to the end of the list.
2. `insert(index, element)`: Adds an element to a specific index.
3. `extend()`: Adds multiple elements to the end of the list.

9. How many methods to delete elements in lists. Total methods:

There are 5 methods to delete elements in lists. They are:

1. `remove()`: Deletes first instance of the element in the list.
2. `del(list[index])`: Deletes the specified index in the list.
3. `del(list)`: Deletes the entire list itself.
3. `pop()`: Deletes the last element of the list.
4. `pop(index)`: Deletes the specified index in the list.
5. `clear()`: Deletes all elements of the list making it empty.

10. How many operators in Python. Write their names and explain:

There are 7 main types of operators in Python. They are:

1. Arithmetic Operators: They do mathematical calculations. Eg: +, -, *, /, %, etc.
2. Assignment Operators: They are used to assign values to variables. Eg: =, +=, -=, etc.